Quantum computing and post-quantum cryptography

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tl;dr

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This could have direct effects on all our lives.

Quantum computing

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This sparked the study of **quantum computing**.

Quantum computers

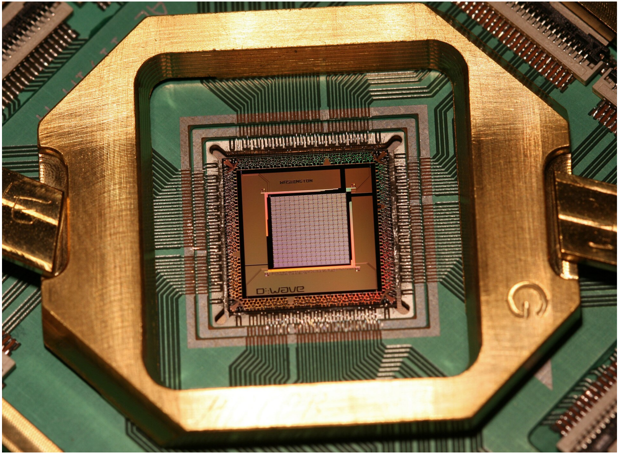

Photo credit: Mwjohnson0, user on Wikipedia

State of the technology

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Hope: Utility-scale quantum computers, quantum advantage

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- Modeling proteins to discover new drugs
- Predicting the weather

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$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Here

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

so $\psi = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ is a vector of length 1 in $\mathbb{C}^2$.

Quantum measurement

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Given an orthonormal basis $\vec{u}, \vec{u}^\perp$ of $\mathbb{C}^2$, measure and observe

- $\vec{u}$ with probability $|\operatorname{proj}_{\vec{u}} \psi|^2$, or
- $\vec{u}^\perp$ with probability $1 - |\operatorname{proj}_{\vec{u}} \psi|^2 = |\operatorname{proj}_{\vec{u}^\perp} \psi|^2$.

Simplest example

Using the orthonormal measurement basis

$$|\vec{u}\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |\vec{u}^{\perp}\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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and measuring the state

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and measuring the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

we observe

- $|0\rangle$ with probability $|\operatorname{proj}_{|0\rangle} \psi|^2 = |\alpha|^2$, or
- $|1\rangle$ with probability $1 - |\operatorname{proj}_{|0\rangle} \psi|^2 = |\operatorname{proj}_{|1\rangle} \psi|^2 = |\beta|^2$.

Need for mathematicians

Qubits are very fragile.

Coding theory detour

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This code uses 3 **physical bits** to encode 1 **logical bit**.

Quantum error correcting

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First time that adding qubits helps accuracy!

Relatedly but separately

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Some gates are "worse" than others for noise.
Circuits must be optimized to minimize their use.

Upshot

And all through the land, all the people rejoice because soon they will solve hard problems thanks to quantum computers!

.... or do they?

Enter the cryptographers

There are those people who love hard problems and wish for them to remain hard.

Keys in cryptography

Algorithms in cryptography have all sorts of functionality.

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These are controlled both by the specific algorithm and a parameter called a **key**.

Two main “kinds” of cryptography

Crypto algorithms fall in two families:

- **Secret** key crypto (or *symmetric* key), and
- **Public** key crypto (or *asymmetric* key).

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To decrypt, use the **same** key k, but choose the kth letter following a given letter in the alphabet.

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Given p, an e.c. $E/\mathbb{F}_p$ and a generator P for $E(\mathbb{F}_p)$,

- a person's **secret key** is a random scalar k, and
- their **public key** is the point $Q = [k]P$.

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In secret/symmetric key crypto, A and B use the **same** secret key.
Therefore the key must remain secret.

In public/asymmetric key crypto, receiver generates **two** keys.
One of them (the encryption key) can be made public.

PKC principle

Public key cryptography is made possible by the fact that there are mathematical operations that are **easy** to do and **hard** to undo.

Quick computational complexity review

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For example, the number of steps to factor a number with n digits grows (almost) exponentially with n.

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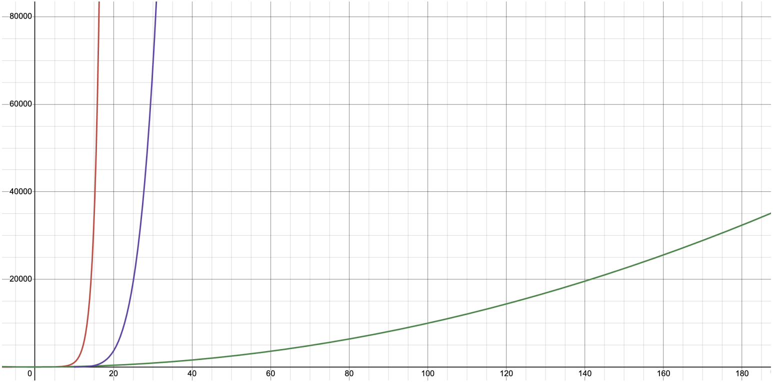
For example, the number of steps to multiply two numbers with n digits grows at most like n^2.

Easy vs hard in picture



Example PKC algorithm: RSA (1978)

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At least it is, until we have quantum computers.

Hardness of problems over time

Newer computers do the same thing but faster.

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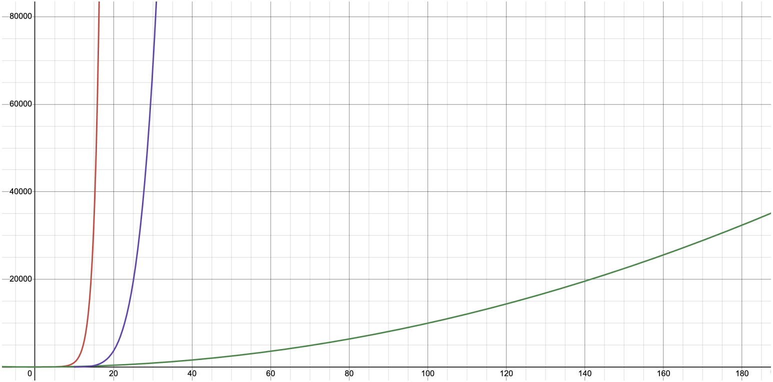
	then	now
RSA-100	few days (1991)	72 mins (2012)
RSA-110	one month (1992)	4 hours (2012)

Specifically, enter Shor

1994: Shor shows how a large-enough quantum computer could factor a number with n digits in n^3 steps.

Easy vs hard picture once more



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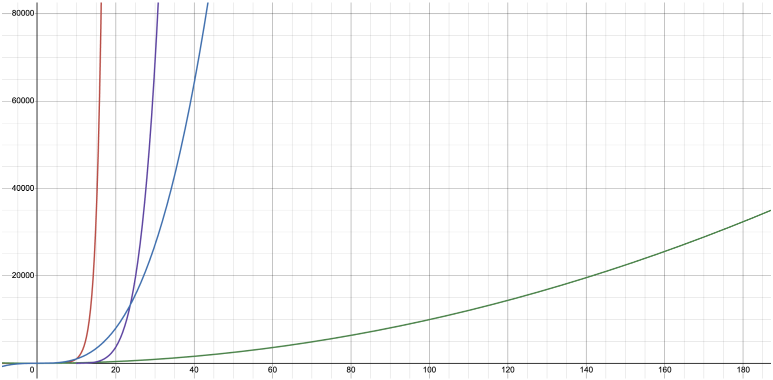

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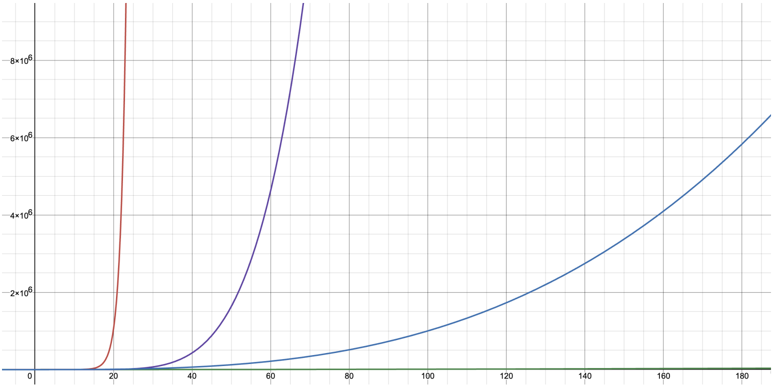

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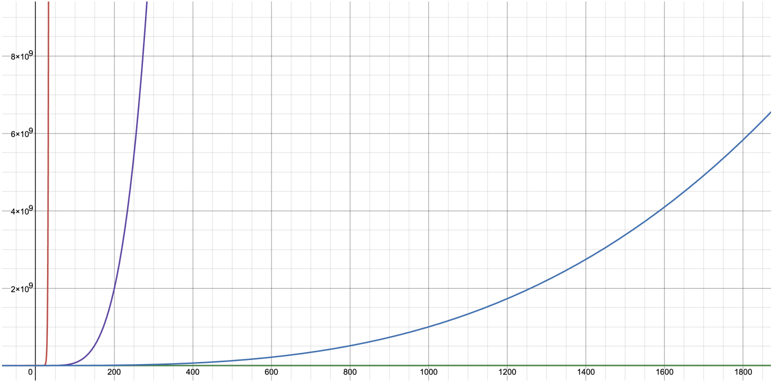

Near-future threat

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Number of operations necessary to factor RSA-230:

Classical:	at least 10^{23} operations
Quantum:	about 3×10^6 operations

Near-future threat?

	Logical qubits	Physical qubits	time IRL
RSA-2048	13,255	102,000	97 days
ECC-256	11,961	26,000	10 days

What to do?

In 2017, NIST begins to **standardize** post-quantum crypto.

Quick note

Don't confuse **post-quantum** crypto with **quantum** crypto!

Post-quantum algorithm families/problems

1. code-based

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5. supersingular elliptic curve isogeny

Issues

Post-quantum cryptography is **heavy**

Algorithm	PK size (bytes)	Ciphertext (bytes)	KeyGen (kcycles)	Encrypt (kcycles)	Decrypt (kcycles)
ECDH	64	64	187	187	187
Kyber-512	800	768	123	155	289
McEliece	261K	96	35K	38	128
HQC	2,249	4,497	87	204	362
BIKE	1,540	1,572	589	97	1,135

Table credit: Angela Robinson at NIST

Issues

Algorithms chosen for standardization based on few problems.

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Competition now in progress to get more signatures.

Need for mathematicians

We need

- more hard problems,
- more algorithms based on them, and
- more people trying to break them.

New hard problems: **Learning With Errors (LWE)**

The hardness of these problems is based on the hardness of finding a short vector in a lattice.

LWE pairs

Fix q a prime, n an integer, and choose a secret vector $\vec{s} \in \mathbb{F}_q^n$.

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Then for $\vec{a} \in \mathbb{F}_q^n$, compute the value

$$b = \vec{a} \cdot \vec{s} + e,$$

for $e \in \mathbb{F}_q$ "small."

We call the pair $(\vec{a}, b)$ an **LWE pair**.

Search LWE

Given m LWE pairs with the same secret $\vec{s}$

$$(\vec{a}_1, b_1), \ldots, (\vec{a}_m, b_m),$$

find the vector $\vec{s}$.

Decision LWE

Given m pairs $(\vec{a}_i, b_i) \in \mathbb{F}_q^n \times \mathbb{F}_q$, determine if the pairs are LWE with some vector $\vec{s}$, or if they were generated at random.

Regev LWE key generation

Fix q, n and $\vec{s}$, and generate m LWE pairs.

The **secret key** is $\vec{s}$ and the **public key** are the LWE pairs.

Regev LWE encryption

To encrypt the bit $m \in \{0, 1\}$, choose a random subset $T \subseteq \{1, 2, \ldots, m\}$ and return the pair $(\vec{a}, b)$, where

$$\vec{a} = \sum_{i \in T} \vec{a}_i$$

and

$$b = \left\lfloor \frac{q}{2} \right\rfloor m + \sum_{i \in T} b_i.$$

Regev LWE decryption

Upon reception of the pair $(\vec{a}, b)$, compute

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If closer to 0 than to $\lfloor \frac{q}{2} \rfloor$, the bit sent was 0 with high probability; if closer to $\lfloor \frac{q}{2} \rfloor$ than to 0, the bit sent was 1 with high probability.

Instead of dot product on $\mathbb{F}_q^n$ (n-dim $\to$ 1-dim),

Lattice-based crypto IRL

Instead of dot product on $\mathbb{F}_q^n$ (n-dim $\rightarrow$ 1-dim),

use a multiplication on $\mathbb{F}_q^n$, so $\vec{a} \cdot \vec{s}$ has dimension n!

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E.g. $K = \mathbb{Q}[\zeta_{2^m}]$, $\mathcal{O}_K = \mathbb{Z}[\zeta_{2^m}]$, $\mathfrak{q}|q$ and $q \equiv 1 \pmod{2^m}$

Thank you!